

ORIGINAL ARTICLE

Generalizability and usefulness of artificial intelligence for skin cancer diagnostics: An algorithm validation study

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Abstract

Background: Artificial intelligence can be trained to outperform dermatologists in image-based skin cancer diagnostics. However, the networks' sensitivity to biases and overfitting may hamper their clinical applicability.

Objectives: The aim of this study was to explain the potential consequences of implementing convolutional neural networks for stand-alone melanoma diagnostics and skin lesion triage.

Methods: In this algorithm validation study on retrospective data, we reproduced and evaluated the performance of state-of-the-art artificial intelligence (convolutional neural networks) for skin cancer diagnostics. The networks were trained on 25,331 annotated dermoscopic skin lesion images from an open-source data set (ISIC-2019) and tested using a novel data set (AISC-2021) consisting of 26,591 annotated dermoscopic skin lesion images. We tested the trained algorithms' ability to generalize to new data and their diagnostic performance in two simulations (melanoma diagnostics and skin lesion triage).

Results: The trained algorithms performed significantly less accurate diagnostics on images of nevi, melanomas and actinic keratoses from the AISC-2021 data set than the ISIC-2019 data set ($p < 0.003$). Almost one-third (31.1%) of the melanomas were misclassified during the melanoma diagnostics simulation, irrespective of their Breslow thickness. Furthermore, the algorithms marked 92.7% of the lesions 'suspicious' during the triage simulation, which yielded a triage sensitivity and specificity of 99.7% and 8.2%, respectively.

Conclusions: Although state-of-the-art artificial intelligence outperforms dermatologists on image-based skin lesion classification within an artificial

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setting, additional data and technological advances are needed before clinical implementation.

KEYWORDS

artificial intelligence, melanoma, skin cancer, skin cancer prevention and early detection

INTRODUCTION

The incidence of melanoma (MEL), the deadliest subtype of skin cancer, is expected to double before 2030.^{1,2} Accurate skin cancer diagnostics based on a clinical examination and dermoscopy, a noninvasive technique, is difficult and requires several years of experience.^{3,4} Early attempts to replace competent clinicians by implementing diagnostic algorithms based on predefined visual criteria were promising, however, none of the methods were appropriately specific, reliable and practical for clinical implementation.^{5–10} This changed in 2017 when convolutional neural networks (CNN), a subtype of artificial intelligence (AI), performed dermatologist-level skin cancer diagnostics.¹¹ Several sequential studies have confirmed that CNNs outperform clinicians in image-based skin cancer diagnostics within a controlled and experimental artificial setting.¹² These results have led to a chain of sensational stories stating that AI has surpassed dermatologists in skin cancer diagnostics.^{13–16} The potential consequences of real-world implementation of CNN algorithms are unclear, but preliminary findings suggest that they would be unreliable and substantially less accurate than clinicians in the clinical evaluation of skin lesions.^{17–20} Unlike humans, algorithms cannot evaluate and treat skin lesions and the accompanying patient from a holistic perspective. Moreover, CNNs are known to generalize poorly towards new data sources and be sensitive towards biases, overfitting and spurious correlations introduced through their training data set.^{4,21–28} Several recent studies suggest that CNNs may provide significant clinical value if used in cooperation with a skilled clinician for narrow tasks such as differentiating between nevi and MELs.^{20,28} However, the media and several commercial companies suggest using AI for patient or clinician-directed automated stand-alone diagnostics of skin lesions. This study aims to measure and explain the generalizability and performance of a state-of-the-art ensemble (group) of CNNs for automated stand-alone skin cancer diagnostics and triage, in a simple and clinically meaningful way.

MATERIALS AND METHODS

In this algorithm validation study, we reproduced an ensemble of CNNs, with dermatologist-level performance in skin cancer diagnostics, hereafter referred to as the international skin image collaboration (ISIC) ensemble. We explored the generalizability of the ISIC ensemble towards new image capturing devices (cameras, dermoscopes, other devices) and its clinical usefulness in MEL diagnostics and skin cancer triage. The ensemble was trained on annotated dermoscopic images from the open-source image repository called ISIC archive. Our group developed a new data set called AISC-2021, which we used during the clinical simulations to test the ensemble's generalizability. We used the continuously maintained open-source data science framework PyTorch.²⁹ The study received an ethical waiver from the Danish Ethics Committee (jr. nr. H-20066667) and was carried out per the Declaration of Helsinki principles. The Danish Health Data Authorities and Data Protection Agency approved access, annotation, anonymization, handling and storage of dermoscopic images (jr. nr. 21/5103 and 18/53664).

Data

This study included two skin lesion image datasets, the ISIC-2019 (ISIC challenge 2019) data set from the ISIC archive and our AISC-2021 data set. Both datasets consist of dermoscopic images annotated with one of the following eight diagnostic labels: actinic keratosis/Bowen's disease (AK), basal cell carcinoma (BCC), benign keratinocytic lesions (BKL), dermatofibroma (DF), MEL, melanocytic nevus (NV), squamous cell carcinoma (SCC) and vascular lesion (VASC).

ISIC-2019 data set

The ISIC-2019 data set consists of a training (ISIC-2019-train) and test (ISIC-2019-test) data set. ISIC-2019-train can be downloaded from the ISIC 2019 challenge website (<https://challenge2019.isic-archive.com>). ISIC-2019-test is

available through an online leaderboard (<https://challenge.isic-archive.com/leaderboards/live>), where anyone can publish and test their algorithms. The ISIC-2019 data set has been thoroughly described elsewhere.^{30,31} For simplicity, we excluded the ISIC-2019 images with an unknown diagnosis (unknown class).

AISC-2021 data set

The AISC-2021 data set consists of 27,638 annotated dermoscopic images of 15,552 pigmented skin lesions from 7122 patients. Images were captured by nurses and doctors at the Department of Dermatology and Allergy Centre, Odense University Hospital, Denmark, from 1 September 2010 to 8 May 2021. All dermoscopic images were photographed using digital dermoscopes (Medicam 800 and 1000, Fotofinder Systems GmbH). Clinical ($n = 9073$) and histopathological ($n = 6479$) diagnoses were manually merged with the dermoscopic images, and the Breslow thickness was added for MELs. Clinical diagnoses consisted of a joint judgement by two to three clinicians, of whom at least one was a board-certified dermatologist. There was no data available on patient ethnicity, Fitzpatrick skin type, image quality, type of dermoscopy (polarised/nonpolarised) or the presence of hair and tattoos. Most dermoscopic images were polarized and photographed following ethanol-moisturization of the skin. Dermoscopic rulers were not present in the dermoscopic images, and ulcers or scars could be found in a minority of the images. All skin lesions in the AISC-2021 data set were registered before excision or as part of a follow-up programme with sequential dermoscopy to rule out malignancy.

We created one training and two test datasets from the AISC-2021 data; AISC-2021-train, AISC-2021-test and AISC-2021-MEL (Melanoma). The AISC-2021-test data set was constructed with the same distribution of diagnoses as the ISIC-2019-test data set, ensuring that we could test the algorithms' generalizability rather than overfitting bias. To prevent data leakage between the training and test datasets, we split data on a lesion level and performed a postsplit analysis of whether images were present in both datasets.

Training and benchmarking of ISIC ensemble

The ISIC ensemble consisted of a group of CNNs trained using a previously described method developed by the winning team (Gessert et al.) in the ISIC challenge in

2019.^{4,30} The ISIC 2019 challenge was an international contest of algorithms for skin lesion diagnostics that attracted scientific and commercial machine learning labs worldwide. We noted overfitting early on, which led us to train the ensemble's CNNs for 60 rather than the 100 epochs originally described by Gessert et al. The term 'epoch' describes the number of times the algorithms are trained on the entire training data set. Furthermore, Gessert et al. supplemented the training data with additional data, which we did not have access to. When the ISIC ensemble was done training, we benchmarked its performance against the original ensemble by Gessert et al. to confirm a successful reproduction.

Generalizability of ISIC ensemble

The generalizability of the ISIC ensemble was tested through a comparison of its performance on two test datasets, one with images from the same sources as the training data (ISIC-2019-test) and another with images from a new data source (AISC-2021-test). To validate whether any potential differences were due to poor generalizability, we trained an identical ensemble of CNNs on a mixed training data set that included images from both data sources (ISIC-2019-train and AISC-2021-train) and tested its (ISIC & AISC ensemble) performance on both test datasets (ISIC-2019-test and AISC-2021-test), see Figure 1.

Clinical simulations

We tested the ISIC ensemble's performance through different interpretations of its probability output in two clinical simulations: MEL diagnostics and skin cancer triage (Figure 2).

MEL diagnostics

The ISIC ensemble's output consisted of probability estimates for each diagnostic class, for example, 80% MEL, 15% NV and 4% DF. We diagnosed each skin lesion during the first simulation based on the diagnostic class that achieved the highest probability estimate, see Figure 3a. All MELs (invasive and in situ) from the AISC-2021 data set (AISC-2021-MEL) were diagnosed in this simulation. The correlation between probability for MEL and Breslow thickness was examined to test whether the ISIC ensemble's accuracy was superior for thick compared to thin MELs.

FIGURE 1 Study design. The ensemble of CNNs was trained using a formerly described method by Gessert et al.³⁰ First, we trained the ensemble on the ISIC-2019-train data set (Step 1) and benchmarked its performance against the original ensemble by Gessert et al. on the ISIC-2019-test data set (Step 2). Second, we explored the generalizability of the ISIC ensemble towards data from the AISC data set. For this purpose, we trained the same ensemble of CNNs on a mixed AISC & ISIC data set (Step 3) and compared its performance against the ISIC ensemble (Step 4). Finally, two clinical simulations evaluated the ISIC ensemble's clinical performance in melanoma diagnostics (AISC-2021-MEL) and skin cancer triage (AISC-2021). CNN, convolutional neural networks; ISIC, international skin image collaboration.

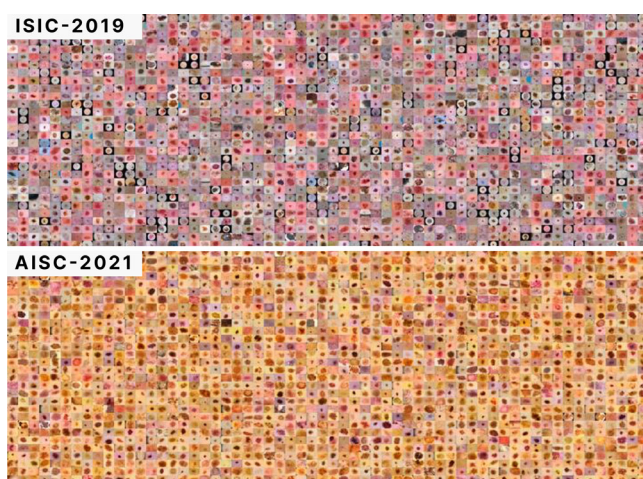
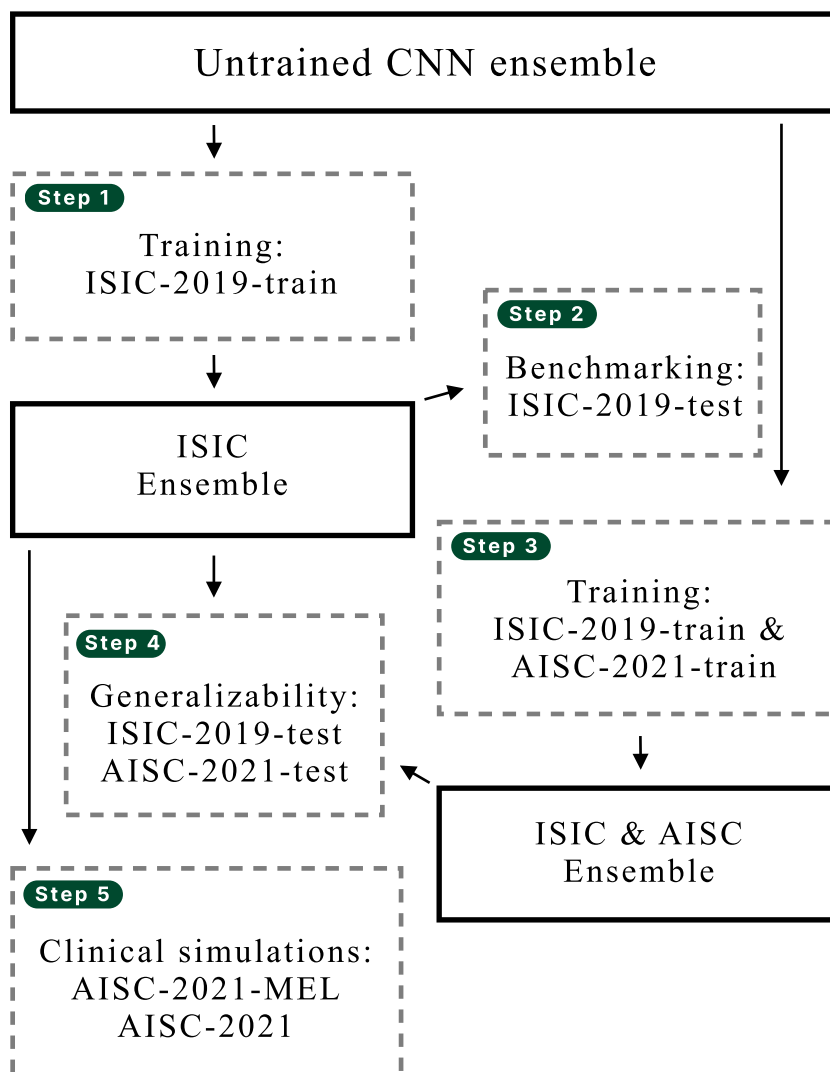


FIGURE 2 ISIC-2019 and AISC-2021 datasets. The figure shows a random sample of 4000 unprocessed miniature dermoscopic images. The upper 2000 images are from the ISIC-2019 data set, while the lower 2000 images are from the AISC-2021 data set. ISIC, international skin image collaboration.

Skin cancer triage

We determined which probability thresholds for BCC, MEL and SCC, that yielded a sensitivity of 99.1% on the ISIC-2019-test data set, see Figure 3b,c. A sensitivity of 99.1% was deemed acceptable as it corresponds to the clinical management sensitivity of an expert dermatologist.³² Lesions were considered suspicious if the probability estimates for MEL, BCC or SCC exceeded their predefined thresholds. We used the entire AISC-2021 data set to evaluate the triage interpretation's sensitivity and specificity.

Statistics

All statistical analyzes were performed in SAS Studio 3.8 (SAS Institute) and Excel 2016 (Microsoft). The

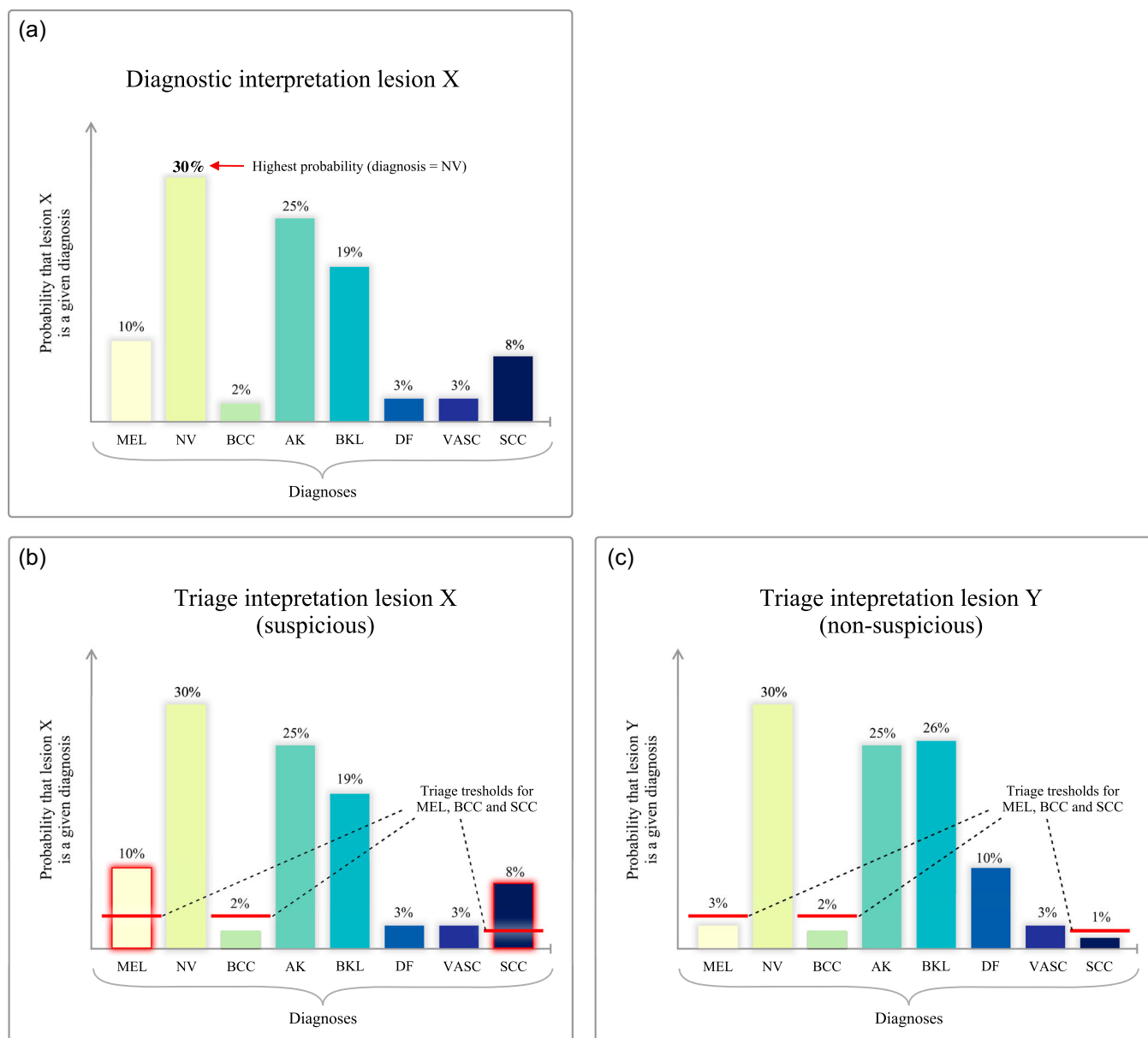


FIGURE 3 CNN diagnostics and triage. When CNN algorithms classify a specific skin lesion, for example, ‘lesion X’ or ‘lesion Y’, they provide probability estimates for each diagnostic category included in the training data set. (a) During the clinical simulation for melanoma diagnostics, skin lesion images were classified as the diagnosis with the highest probability among the eight diagnostic categories. For example, lesion X would be diagnosed as NV since this category achieved the highest probability score (30%). (b) During the triage simulation, skin lesions were classified as suspicious if the probability for MEL, BCC or SCC surpassed predefined acceptable thresholds (red lines). This means that lesion X would be considered suspicious for malignancy as its probability for MEL and SCC are higher than the acceptable thresholds. (c) Lesion Y, on the other hand, is not suspicious for malignancy as its probability for MEL, BCC and SCC does not surpass the thresholds. BCC, basal cell carcinoma; CNN, convolutional neural networks; MEL, melanoma; NV, nevus; SCC, squamous cell carcinoma.

areas under the receiver operating characteristic curves (ROC AUC) were compared using the Hanley–McNeil test,³³ and the correlation between Breslow thickness and MEL probability was evaluated using Pearson's correlation test. *p* Values below 0.05 were considered significant (Table 1).

RESULTS

Benchmarking and generalizability

Our ensemble's (ISIC ensemble) performance on the ISIC-2019-test was comparable (Δ ROC AUC = 0.004,

TABLE 1 Class (diagnosis) distribution across the training and test datasets

Class	ISIC-2019-train	ISIC-2019-test ^a	AISC-2021-train	AISC-2021-test	AISC-2021-MEL
MEL	4522 (18%)	1318 (20%)	954 (4%)	408 (20%)	1115 (100%)
NV	12,875 (51%)	2636 (40%)	21,774 (89%)	805 (40%)	-
BCC	3323 (13%)	1071 (16%)	211 (1%)	326 (16%)	-
AK	867 (3%)	412 (6%)	63 (<1%)	127 (6%)	-
BKL	2624 (10%)	824 (13%)	1088 (4%)	216 (11%)	-
DF	239 (1%)	82 (1%)	161 (1%)	31 (2%)	-
VASC	253 (1%)	82 (1%)	330 (1%)	35 (2%)	-
SCC	628 (2%)	165 (3%)	9 (<1%)	53 (3%)	-
Total	25,331	6590	24,590	2001	1115

Abbreviations: AK, actinic keratosis; BCC, basal cell carcinoma; BKL, benign keratinocytic lesion; DF, dermatofibroma; ISIC, international skin image collaboration; MEL, melanoma; NV, nevus; SCC, squamous cell carcinoma; VASC, vascular lesions.

^aApproximations based on a percentage value extracted from the ISIC-2019 leaderboard.

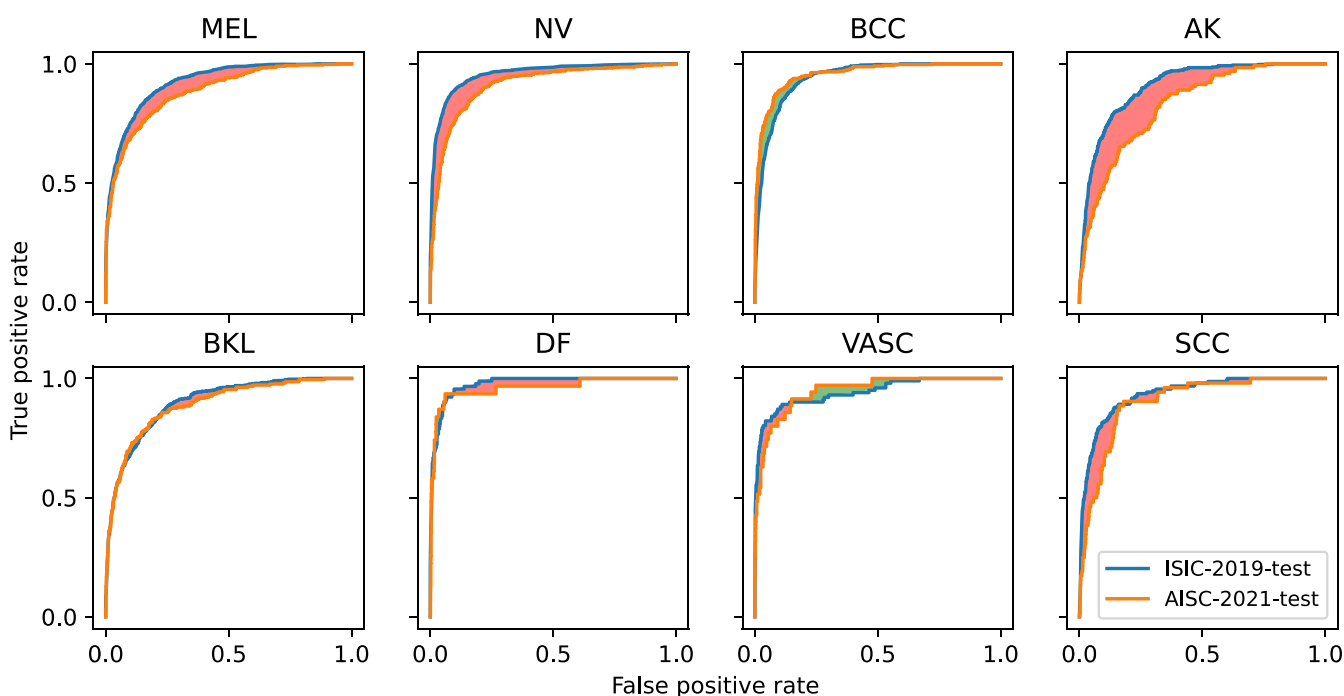


FIGURE 4 ISIC ensemble's generalizability. The figure shows the ISIC ensemble's ROC curves for each diagnosis category on the ISIC-2019-test and AISC-2021-test datasets. Blue and orange lines are the ISIC ensemble's performance on the ISIC-2019-test (blue) and AISC-2021-test (orange) datasets, respectively. Red areas represent a difference between the ROC curves when the ISIC-2019-test curve is superior. Conversely, green areas represent the difference when the ISIC-2019-test curve is inferior. ISIC, international skin image collaboration; ROC, receiver operating characteristic.

$p = 0.059$) to that of the original ensemble by Gessert et al.³⁰ The ensemble's ROC AUCs for NV, MEL and AK were significantly lower on the AISC-2021-test compared to the ISIC-2019-test data set, see Figure 4 and Table 2. The combined ISIC & AISC ensemble had a markedly better performance than the ISIC

ensemble on both test datasets (ISIC-2019-test and AISC-2021-test). In fact, the ISIC & AISC ensemble exhibited signs of AISC-2021 overfitting, with a disproportionately large improvement on the AISC-2021-test compared to the ISIC-2019-test, see Table 2.

MEL diagnostics

The ISIC ensemble had a 68.9% MEL sensitivity and thus misdiagnosed 31.1% of the MELs (in situ and invasive).

TABLE 2 Performance on test datasets

Class	ISIC ensemble			ISIC & AISC ensemble		
	ROC AUC ISIC-2019-test	ROC AUC ASIC-2021-test	P Value	ROC AUC ISIC-2019-test	ROC AUC AISC-2021-test	P Value
MEL	0.923	0.886	0.003	0.929	0.944	0.108
NV	0.956	0.913	<0.001	0.953	0.940	0.039
BCC	0.941	0.956	0.118	0.943	<u>0.978</u>	<0.001
AK	0.908	0.830	0.002	0.911	0.921	0.589
BKL	0.898	0.891	0.638	0.906	<u>0.941</u>	0.011
DF	0.977	0.960	0.541	0.980	0.987	0.706
VASC	0.942	0.946	0.884	0.945	<u>0.999</u>	<0.001
SCC	0.935	0.901	0.285	0.931	0.940	0.744

Note: The table shows both ensembles' ROC AUCs for each class (diagnosis category) on the ISIC-2019-test and AISC-2021-test datasets. ROC AUCs on the AISC-2021-test have been highlighted if they are significantly worse (italic) or better (underline) than the ISIC-2019-test ROC AUCs. Bold p values are significant ($p < 0.05$).

Abbreviations: AK, actinic keratosis; BCC, basal cell carcinoma; BKL, benign keratinocytic lesion; DF, dermatofibroma; ISIC, international skin image collaboration; MEL, melanoma; NV, nevus; ROC AUCs, areas under the receiver operating characteristic curves; SCC, squamous cell carcinoma; VASC, vascular lesions.

There was no significant correlation between the Breslow thickness of MELs and the probability of MEL provided by the ensemble ($r = 0.06$, $p = 0.12$), see Figure 5.

Skin cancer triage

With predefined probability thresholds of 3.6%, 3.6% and 0.6% for MEL, BCC and SCC, the ensemble categorized 1138 (7.3%) of the AISC-2021 lesions as nonsuspicious and the remaining 14,413 (92.7%) lesions as suspicious for malignancy. Five (0.4%) of the dismissed lesions were malignant, and 1687 (11.7%) of the suspicious lesions were MEL, BCC or SCC. The five incorrectly dismissed malignant tumours were four BCCs and one SCC. The triage simulation yielded a sensitivity and specificity of 99.7% and 8.2%, respectively.

DISCUSSION

This study evaluated and visualized two critical pitfalls concerning CNN ensembles for skin cancer diagnostics, namely the generalizability and sensitivity-specificity trade-off. We found that a state-of-the-art ensemble of CNNs with dermatologist-level performance within a highly controlled experimental setting generalized poorly towards images of nevi, MELs and actinic keratoses that had been captured using different image capturing devices (cameras and dermoscopes) than the training data set. The ensemble misclassified 31.1% of all MELs during the diagnostic simulation, irrespective of their

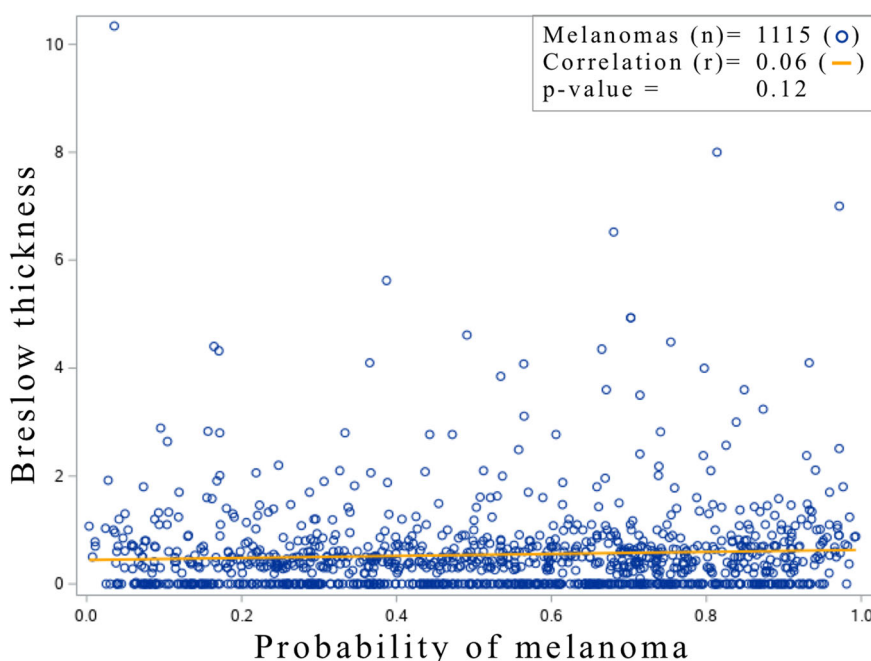


FIGURE 5 Breslow thickness and probability of melanoma. Scatter plot depicting the Breslow thickness (y-axis) and probability of melanoma (x-axis) for all 1115 melanomas (blue circles) in the AISC-2021-MEL data set. The orange line in the lower part of the plot represents Pearson's correlation line ($r = 0.06$, $p = 0.12$). CNN, convolutional neural networks; ISIC, international skin image collaboration; MEL, melanoma.

Breslow thickness. Moreover, the ensemble categorized less than a tenth (7.3%) of the 15,552 AISC-2021 lesions as nonsuspicious during the triage simulation. In other words, the algorithms would be useless for stand-alone automated diagnostics and triage in the hands of untrained operators, as this would result in a large number of dismissed MELs (diagnostics) or referred benign skin lesions (triage).

The ISIC ensemble's generalizability towards the AISC-2021 data was poor, despite being trained on a deliberately heterogeneous training data set (ISIC-2019-train) with manually preprocessed images from multiple skin cancer clinics and departments.^{30,31} The algorithms' performance across both test datasets improved significantly when AISC-2021 data was included in the training data set, although the algorithms overfitted towards the AISC-2021 data set. Our results were confirmed in a recent publication by Combalia et al. that examined the generalizability of the top three algorithms from the ISIC 2019 challenge. The diagnostic accuracy of the algorithms fell by more than 20% when they were asked to diagnose skin lesions from new sources.²⁸ These results are significant as systematic errors caused by the algorithms' inability to generalize could result in fatal patient outcomes if doctors and nurses become wholly reliant on the algorithm's predictions. Related work from cardiology has shown that clinicians quickly forget how to interpret electrocardiograms when automated interpretations are implemented.^{34,35} Similarly, Tschandl and colleagues recently showed that inexperienced clinicians blindly trust diagnostic algorithms for skin cancer diagnostics.²⁰ We tested the algorithms' generalizability using two test datasets that were almost identical except for the devices used to capture the dermoscopic images. In a natural clinical setting, similar subtle colour differences could easily be caused by faltering dermoscopic diodes, new dermoscopes, new camera devices or camera-software updates.

The generalizability of CNN algorithms can be improved by increasing the size and heterogeneity of the training datasets, thereby encompassing all patient demographics, diagnoses and image capturing modalities encountered within the intended clinical setting.²⁵ The ISIC community strives to provide a heterogeneous data set through the continuously growing open-source ISIC-archive image repository. However, there is no guarantee that the ISIC-archive or similar datasets will ever be large or heterogeneous enough to support the development of genuinely reliable diagnostic algorithms for the rapidly evolving clinical ecosystem. For this reason, we suggest that diagnostic algorithms should be prospectively

validated for the intended setting before clinical implementation and continuously after that at pre-determined intervals to ensure rapid identification and mitigation of shifts between input data and the algorithms' training data set. Alternatively, the method and devices used for capturing clinical images should be standardized, ensuring perfect alignment between the input data and the algorithms' training data.³⁶ The safety margins would be microscopic for approaches that rely on standardization as the overfitted algorithms' accuracy would drop markedly when minor technical issues or human errors are introduced.

CNN algorithms perform more accurate skin cancer diagnostics than dermatologists within an artificial and highly controlled test setting.¹² However, in real-life clinicians collect additional diagnostic data, confer with colleagues, act cautiously when in doubt and follow up on suspicious skin lesions. In this study, the 'superior' algorithms misclassified 347 (31.1%) of the 1115 MELs, all diagnosed and treated by the clinicians at the Department of Dermatology and Allergy Center, Odense University Hospital.

For this study, we intentionally chose a very safe triage threshold (sensitivity: >99.1%), corresponding to that of an expert dermatologist with all standard treatment and follow-up options at their disposal. The simulations represented a clinical scenario where the operator of the algorithm had little or no experience in skin cancer diagnostics, for example, a primary care setting. There were no external safety mechanisms, such as the oversight of a domain expert, within the simulated clinical setting, and we, therefore, prioritized safety rather than effectivity. As planned, the algorithms provided safe (sensitivity 99.7%) but ineffective (8.2% specificity) skin cancer triage. The specificity was unacceptably low for the intended unspecialized clinical setting. If implemented, it is likely that the algorithms would drive an increased number of unnecessary benign dermatological referrals, further burdening an already strained speciality.³⁷ When technological solutions such as teledermatology reduce the effort related to dermatological second opinions for skin lesions, primary care providers request significantly more dermatological consultations.³⁸ The clinical usefulness of a binary 'suspicious' or 'nonsuspicious' output is questionable, as the various subtypes of skin conditions require different management, treatment and patient communication strategies.^{39–41} Finally, there is little educational value in a triage algorithm that marks 92.7% of all skin lesions 'suspicious'.

Many MELs (in situ and invasive) are potentially overdiagnosed benign or low-grade malignant melanocytic

lesions,^{42,43} and this may also be the case for some of the AISC-2021 MELs. However, the Breslow thickness is a good prognostic marker for MEL mortality and recurrence, and we, therefore, consider it an inverse proxy for overdiagnosis.⁴⁴ There was no correlation between the MELs' Breslow thickness and the ISIC ensemble's probability estimates for MEL, suggesting that MELs were misclassified irrespective of whether they were potentially overdiagnosed or not.

This study had several limitations. First, we attempted to simulate the clinical implementation of automated algorithms for MEL diagnostics and skin cancer triage in a primary care setting. However, our simulation data set (AISC-2021) came from a specialized department for skin cancer diagnostics where all registered lesions were interesting or suspicious enough to warrant an image registration. The AISC-2021 data set is heavily skewed towards malignant and atypical lesions, similar to the ISIC-2019 data set. We theorize that the ISIC ensemble would perform significantly worse in a primary care setting where most lesions are benign. Second, the average difficulty of the AISC-2021 data set is unknown as there were no human raters in this study. The AISC-2021 data set included more benign diagnoses than the ISIC-2019, which may indicate that it was easier than the ISIC-2019 data set. Third, this study investigated stand-alone automated CNN diagnostics in the hands of inexperienced clinicians. However, CNNs for clinical assistance among skilled clinicians and augmenting the training of junior clinicians seems to hold a far greater clinical potential.²⁰

To summarize, this study examined the generalizability and clinical performance of a formerly described state-of-the-art CNN ensemble for skin cancer diagnostics. The algorithms generalized poorly towards new image formats and performed significantly worse MEL diagnostics and skin cancer triage than clinicians.

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The patients in this manuscript have given written informed consent to the publication of their case details. There are no patients involved in this publication as all skin lesion images were anonymized before inclusion in this study. However, the inclusion of the previous sentence is a requirement by the Journal of the European Academy of Dermatology and Venerology. Grant 9093-00005B (Innovation Fund Denmark).

CONFLICTS OF INTEREST

Niels K. Ternov: CEO of Melatech ApS, a software company developing a platform for clinical management of skin lesions and digital education in skin cancer diagnostics. The remaining authors declare that there are no conflicts of interest.

DATA AVAILABILITY STATEMENT

The ISIC-2019 datasets are openly available in through the ISIC archive and ISIC 2019 leaderboard at <https://challenge2019.isic-archive.com> and <https://challenge.isic-archive.com/leaderboards/live>. The AISC-2021 datasets are available from the corresponding author upon reasonable request. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

ETHICS STATEMENT

The patients in this manuscript have given written informed consent to publication of their case details. The study received an ethical waiver from the Danish Ethics Committee (jr. nr. H-20066667) and was carried out per the Declaration of Helsinki principles. The Danish Health Data Authorities and Data Protection Agency approved access, annotation, anonymization, handling, and storage of the dermoscopic images (jr. nr. 21/5103 and 18/53664).

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